RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
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Question Paper ID: 190663

Course Code: CD252IA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

COMMON TO AIML / CD / CS / CY / IS

Time : 3 Hours

Database Management Systems

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10

Part A

Question No	Question	M	CO	BT
1.1	Define database instance.	01	2	1
1.2	Differentiate between sophisticated, and stand-alone end-users.	02	2	2
1.3	Write the difference between DELETE and TRUNCATE?	02	2	2
1.4	What are the two types of structural constraints?	01	1	1
1.5	Why is an entity called a weak entity type?	02	2	2
1.6	Define a minimal set of functional dependencies.	02	1	1
1.7	Why are aggregates important in NoSQL?	02	2	2
1.8	What is the "Data Locality" principle in Hadoop?	02	1	1
1.9	Explain Relationship Types and Relationship Sets with examples.	02	2	2
1.10	List the types of relationships based on degree.	02	1	2
1.11	What are serial Schedules? Give example.	02	1	2

Part B

Question No	Question	M	CO	BT
2a	Write the ER diagram for the Car Rental System , providing a clear visualization of how different components interact within the system. Employee (both regular workers and supervisors), Vehicle (all available cars in the rental fleet), and Customer (who can either rent a vehicle	10	2	3

	directly or make a reservation).Supervisors have a management role over other employees and handle customer reservations. When a customer rents a vehicle, they provide complete information and full payment. For reservations, customers specify pickup dates and other details. The Key entities are Employee, Reservation, Customer, Rent , Vehicle. Identify the relationship existing between them, Participation constraints and mapping. Specify any assumption you make.			
2b	Design, document, and justify a complete ER model for an enterprise-level system of your choice, explicitly addressing relationship types, roles, structural constraints, and weak entity types. Critically evaluate its strengths and limitations.	06	3	6
3a	Explain the concept of cardinality ratio in an ER model. Describe the different types of cardinality ratios with suitable examples.	04	1	3
3b	Describe the Division (÷) operator in Relational Algebra and illustrate its working using an example.	04	2	2
3c	Consider the following relational schema for a Hospital Management System: • DOCTOR (DId, DName, Specialty, Charge) • PATIENT (PId, PName, Age, City) • TEST (TId, TName, Cost) • REPORT (PId, TId, DId, Date) (Note: The REPORT table links Patients, Tests, and Doctors, signifying which doctor performed which test on which patient.) Write the Relational Algebra expressions for the following queries (2 Marks each): i). Retrieve the names of all Doctors who specialize in 'Neurology' and charge more than 1500. ii). Find the names of Patients who have undergone the test named 'MRI Scan'. iii). Find the DId of doctors who have performed either a 'CT Scan' or an 'X-Ray' (or both). iv). Find the names of Patients who have undergone all the tests available in the TEST table.	08	3	4
	OR			
4a	Consider the following "Sailors and Boats" schema. Write Relational Algebra queries for the following. Sailors (sid, sname, rating, age) Boats (bid, bname, color) Reserves (sid, bid, day)	10	3	4

i). Find the names of sailors who have reserved a red boat.

ii). Find the names of sailors who have reserved at least one boat.

iii). Find the colors of boats reserved by 'Lubber'.

iv). Find the names of sailors who have reserved a Red boat OR a Green boat.

v). Find the names of sailors who have reserved ALL boats.

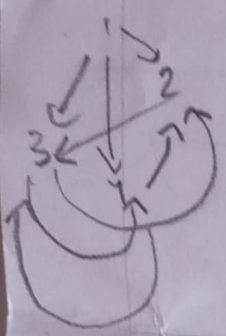
4b	Compare and contrast Relational Algebra Operations from Set Theory (UNION, INTERSECTION, MINUS) with examples.	06	3	3
5a	Consider the relation R=(A1,A2,A3,A4,A5) and the set of functional dependencies FD={A1→A3 A5," " A5→A1 A4," " A3 A4→A_2}. Determine whether the decomposition D={R1,R2,R3}, where R_1={A1,A2,A3,A5}, R_2={A1,A3,A4}, and R_3={A4,A5}, is a lossless join decomposition. Show all the necessary steps as per the algorithm	10	3	5
5b	What is equivalence of functional dependency sets? Explain how equivalence is tested.	06	3	2

OR

6a	An e-commerce system maintains: CUSTOMER (CustID, CustName, Email) ORDERS (OrderID, CustID, OrderDate, Amount) Write SQL queries to i) Display customer names with order details. ii) Find total order amount per customer. iii) Display customers who have placed more than one order. iv) Retrieve orders placed in the last 6 months. v) Find the highest order amount.	10	3	3
6b	Given a relation R(A, B, C, D) with the following sets of functional dependencies: $F_1 = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$ $F_2 = \{ A \rightarrow C, C \rightarrow D, A \rightarrow B \}$ Determine whether F_1 and F_2 are equivalent.	06	3	4

7a	Check weather given schedules are serial/non-serial/serializable/non serializable/conflict serializable schedules. Justify your answer.	10	3	4

Schedule S1	Schedule S2	Schedule S3	Schedule S4
R1(A)	R1(A)	R1(A)	R1(A)
W1(A)	R2(A)	R2(A)	W1(A)
R1(B)	W1(A)	W2(A)	R2(A)
W1(B)	W2(A)	W1(A)	W2(A)
R2(A)	R1(B)	R1(B)	R2(B)
W2(A)	R2(B)	R2(B)	W2(B)
R2(B)	W1(B)	W1(B)	R1(B)
W2(B)	W2(B)	W2(B)	W1(B)



7b	Compare binary locks to exclusive/shared locks. Why is the latter type of locks preferable?	06	2	2

OR

8a	Differentiate between shared locks and exclusive locks with suitable examples.	04	2	2
8b	Analyze the structure and role of the system lock table in DBMS concurrency control. Illustrate its components and significance.	08	3	4
8c	Explain the basic idea of Two-Phase Locking with an example.	04	1	2
9a	List out the key features of schemaless databases.	04	2	2
9b	Explain the Hadoop ecosystem and the role of HDFS and MapReduce in it.	04	3	2
9c	What is the architectural function of YARN within the Hadoop ecosystem, and how does it manage resource allocation and job scheduling across a distributed cluster?	08	3	4

OR

10a	Describe the end-to-end execution lifecycle of a MapReduce job, detailing the transition of data through the InputSplitting, Mapping, Shuffling/Sorting, and Reducing phases.	10	3	2
10b	Describe Peer-to-Peer Replication in databases. Explain its working mechanism with a diagram and highlight its key advantages.	06	1	2